

Summation of Series.

Ct's method

Summation based on geometric Series

$$e^{i\theta} = \cos\theta + i\sin\theta$$

$$e^{-i\theta} = \cos\theta - i\sin\theta$$

$$e^{in\theta} = \cos n\theta + i\sin n\theta$$

$$a + ar + ar^2 + \dots = a/1-r$$

$$\underbrace{a + ar + ar^2 + \dots + ar^{n-1}}_{n \text{ terms}} = \begin{cases} \frac{a(r^n - 1)}{r - 1}, & |r| > 1 \\ \frac{a(1 - r^n)}{1 - r}, & |r| < 1 \end{cases}$$

Q. Sum to n terms and sum to infinity for the following series? $1 + a \cos\theta + a^2 \cos 2\theta + a^3 \cos 3\theta + \dots$, $|a| < 1$

$$\text{Let } C = 1 + a \cos\theta + a^2 \cos 2\theta + a^3 \cos 3\theta + \dots + a^{n-1} \cos(n-1)\theta$$

$$S = a \sin\theta + a^2 \sin 2\theta + \dots + a^{n-1} \sin(n-1)\theta$$

$$e^{i\theta} S = 1 + a(\cos\theta + i\sin\theta) + a^2(\cos 2\theta + i\sin 2\theta) + \dots + a^{n-1}(\cos(n-1)\theta + i\sin(n-1)\theta)$$

$$= 1 + a e^{i\theta} + a^2 e^{i2\theta} + \dots + a^{n-1} e^{i(n-1)\theta}$$

$$= 1 + a e^{i\theta} + (a e^{i\theta})^2 + \dots + (a e^{i\theta})^{n-1}$$

Since $|a| < 1$

$$= \frac{1 - (a e^{i\theta})^n}{1 - a e^{i\theta}}$$

$$= \frac{1 - (a e^{i\theta})^n}{1 - a e^{i\theta}} \times \frac{(1 - a e^{-i\theta})}{1 - a e^{-i\theta}}$$

$$= \frac{1 - ae^{-i\theta} - (ae^{i\theta})^n - a^{n+1} \frac{e^{in\theta} - e^{-i\theta}}{e^{i\theta} - e^{-i\theta}}}{1 - ae^{-i\theta} + a^2 e^{i\theta} - a^2 e^{-i\theta} - ae^{i\theta}}$$

$$e + is = \frac{1 - a(\cos\theta - i\sin\theta) - a^n(\cos n\theta + i\sin n\theta) + a^{n+1}(\cos(n-1)\theta + i\sin(n-1)\theta)}{1 - a(\cos\theta - i\sin\theta) + a^2(\cos\theta + i\sin\theta) + a^2}$$

$$e = \frac{1 - a\cos\theta - a^n\cos n\theta + a^{n+1}\cos(n-1)\theta}{1 - 2a\cos\theta + a^2}$$

As $n \rightarrow \infty$, since $|a| < 1$, $a^n > 0$, $a^{n+1} \rightarrow 0$

$$C_\alpha = \frac{1 - a\cos\theta}{1 - 2a\cos\theta + a^2}$$

Q. Sum of the series $\sin\alpha + \frac{1}{2}\sin 2\alpha + \frac{1}{2^2}\sin 3\alpha + \dots$

$$\text{Let } S = \sin\alpha + \frac{1}{2}\sin 2\alpha + \frac{1}{2^2}\sin 3\alpha + \dots$$

$$e = \cos\alpha + \frac{1}{2}\cos 2\alpha + \frac{1}{2^2}\cos 3\alpha + \dots$$

$$e + is = (\cos\alpha + i\sin\alpha) + \frac{1}{2}(\cos 2\alpha + i\sin 2\alpha) + \dots$$

$$= e^{i\alpha} + \frac{1}{2}e^{i2\alpha} + \frac{1}{2^2}e^{i3\alpha} + \dots$$

$$= \frac{e^{i\alpha}}{1 - e^{i\alpha/2}}$$

$$= \frac{2e^{i\alpha}}{2-e^{i\alpha}} \times \frac{2-e^{-i\alpha}}{2-e^{-i\alpha}}$$

$$= \frac{4e^{i\alpha} - 2e^{i\alpha}e^{-i\alpha}}{4-2e^{i\alpha}-2e^{-i\alpha}+1}$$

$$= \frac{4e^{i\alpha} - 2}{5-2(e^{i\alpha}+e^{-i\alpha})}$$

$$C+iS = \frac{4(\cos\alpha + i\sin\alpha) - 2}{5-2(\cos\alpha + i\sin\alpha)} = \frac{4\cos\alpha + 4i\sin\alpha - 2}{5-2\cos\alpha - 2i\sin\alpha}$$

$$\Rightarrow S = \frac{4\sin\alpha}{5-4\cos\alpha}$$

Q. Sum the series. $\cos\alpha, \cos\alpha + \cos^2\alpha, \cos 2\alpha + \cos^3\alpha, \cos 3\alpha + \dots$

$$\text{Let } C = \cos\alpha, \cos\alpha + \cos^2\alpha, \cos 2\alpha + \cos^3\alpha, \cos 3\alpha + \dots$$

$$S = \cos\alpha, \sin\alpha + \cos^2\alpha, \sin 2\alpha + \cos^3\alpha, \sin 3\alpha + \dots$$

$$C+iS = \cos\alpha(\cos\alpha + i\sin\alpha) + \cos^2\alpha(\cos 2\alpha + i\sin 2\alpha) + \dots$$

$$= \cos\alpha \cdot e^{i\alpha} + \cos^2\alpha \cdot e^{i2\alpha} + \dots$$

$$= \frac{\cos\alpha \cdot e^{i\alpha}}{1 - \cos\alpha \cdot e^{i\alpha}}$$

$$= \frac{\cos \alpha \cdot e^{i\alpha}}{1 - \cos \alpha \cdot e^{i\alpha}} \times \frac{1 - \cos \alpha \cdot e^{-i\alpha}}{1 - \cos \alpha \cdot e^{-i\alpha}}$$

$$= \frac{\cos \alpha \cdot e^{i\alpha} - \cos^2 \alpha}{1 - \cos \alpha \cdot e^{-i\alpha} - \cos \alpha \cdot e^{i\alpha} + \cos^2 \alpha}$$

$$= \frac{\cos \alpha (e^{i\alpha} - \cos \alpha)}{1 - \cos \alpha (e^{i\alpha} + e^{-i\alpha}) + \cos^2 \alpha}$$

$$= \frac{\cos \alpha (\cos \alpha + i \sin \alpha) - \cos^2 \alpha}{1 - \cos \alpha \cdot 2 \cos \alpha + \cos^2 \alpha}$$

$$= \frac{\cos^2 \alpha + i \cos \alpha \sin \alpha - \cos^2 \alpha}{1 - \cos^2 \alpha}$$

$$= \frac{i \cos \alpha \sin \alpha}{\sin^2 \alpha} = \frac{i \cos \alpha}{\sin \alpha} = i \cot \alpha$$

$$\therefore C + iS = i \cot \alpha$$

Since there is no real part, we've C=0

HW.
Q. Sum the series

$$1. \sin \alpha \cdot \sin \alpha + \sin^2 \alpha \cdot \sin \alpha + \sin^3 \alpha \cdot \sin \alpha + \dots$$

$$2. \sin \alpha \cos \alpha + \sin^2 \alpha \cdot \cos^2 \alpha + \sin^3 \alpha \cos^3 \alpha + \dots$$

$$Q. \sin \alpha + c (\sin(\alpha+\beta)) + c^2 \sin(\alpha+2\beta) + c^3 \sin(\alpha+3\beta) + \dots$$

$$\text{Let } S = \sin \alpha + c \sin(\alpha+\beta) + c^2 \sin(\alpha+2\beta) + \dots$$

$$C = \cos \alpha + c \cos(\alpha+\beta) + c^2 \cos(\alpha+2\beta) + \dots$$

$$C + iS = (\cos \alpha + i \sin \alpha) + c (\cos(\alpha+\beta) + i \sin(\alpha+\beta)) + c^2 [\cos(\alpha+2\beta) + i \sin(\alpha+2\beta)] + \dots$$

$$= e^{i\alpha} + c \cdot e^{i(\alpha+\beta)} + c^2 \cdot e^{i(\alpha+2\beta)} + \dots$$

$$= \frac{e^{i\alpha}}{1 - ce^{i\beta}}$$

$$= \frac{e^{i\alpha}}{1 - ce^{i\beta}} \times \frac{1 - ce^{-i\beta}}{1 - ce^{-i\beta}}$$

$$= \frac{e^{i\alpha} - ce^{i(\alpha-\beta)}}{1 - ce^{-i\beta} - ce^{i\beta} + c^2}$$

$$= \frac{e^{i\alpha} (\cos \alpha + i \sin \alpha) - c (\cos(\alpha-\beta) + i \sin(\alpha-\beta))}{1 - c(e^{i\beta} + e^{-i\beta}) + c^2}$$

$$= \frac{\cos \alpha - c \cos(\alpha-\beta) + i(\sin \alpha - \sin(\alpha-\beta))}{1 - 2c \cos \beta + c^2}$$

$$\therefore S = \sin \alpha - c \sin(\alpha-\beta) / 1 - 2c \cos \beta + c^2$$